ALGEBRA 1 · PERSONALIZED PRACTICE

# Math Practice

2026-11-01 · 20 questions

## I CAN DISTINGUISH FUNCTIONS FROM RELATIONSHIPS THAT ARE NOT FUNCTIONS.

**1.** I can distinguish functions from relationships that are not functions.

Pairs $(1, 5), (2, 5), (3, 5)$ all have output 5. Is this a function? Explain.

**2.** I can distinguish functions from relationships that are not functions.

Does $\{(1, 2), (1, 3), (2, 4)\}$ represent a function? Explain.

**3.** I can distinguish functions from relationships that are not functions.

**Identify the repeated input.**

| x | -1 | 2 | -1 |
|---|---|---|---|
| y | 3 | 5 | 8 |

**4.** I can distinguish functions from relationships that are not functions.

Is $(1, 3), (2, 3), (3, 3)$ a function? Evaluate $2(-3) + 5$.

## I CAN IDENTIFY THE DOMAIN AND RANGE OF A RELATION.

**5.** I can identify the domain and range of a relation.

Tickets cost $6 each, and at most 8 tickets may be bought. What inputs are reasonable for the total-cost rule?

**6.** I can identify the domain and range of a relation.

Find the range of $\{(1, 4), (2, 4), (3, -1)\}$.

**7.** I can identify the domain and range of a relation.

Find the domain of $\{(-3, 1), (0, 4), (2, 7)\}$.

**8.** I can identify the domain and range of a relation.

The function $f(x) = \sqrt{x}$ has real outputs. Explain why negative inputs are excluded.

## I CAN CONNECT LINEAR GROWTH IN TABLES, GRAPHS, AND EQUATIONS.

**9.** I can connect linear growth in tables, graphs, and equations.

**Complete the table.**

| x | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| y | 4 | 7 | 10 | ? |

**10.** I can connect linear growth in tables, graphs, and equations.

**Write an equation.**

Slope 4; y-intercept −2.

**11.** I can connect linear growth in tables, graphs, and equations.

The outputs 2, 5, 8, 11 increase by a constant amount. What is the change per step?

**12.** I can connect linear growth in tables, graphs, and equations.

| x | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| A | 1 | 4 | 7 | 10 |
| B | 1 | 4 | 9 | 16 |

Which is linear? Give numerical evidence.

**I CAN CALCULATE AND INTERPRET SLOPE AS A CONSTANT RATE OF CHANGE.**

**13.** I can calculate and interpret slope as a constant rate of change.

One line has slope −2 and another has slope 3. Compare their directions and rates.

**14.** I can calculate and interpret slope as a constant rate of change.

**Find the slope.**

(−2,1), (2,9)

**15.** I can calculate and interpret slope as a constant rate of change.

Find the slope through (0,3) and (4,11). Then evaluate $f(2)$ for $f(x) = 3x - 1$.

**16.** I can calculate and interpret slope as a constant rate of change.

Find the slope through $(0, 2)$ and $(3, 8)$.

## I CAN COMPARE VERTICAL AND HORIZONTAL CHANGES TO FIND SLOPE.

**17.** I can compare vertical and horizontal changes to find slope.

Find rise/run from $(-2,1)$ to $(2,-3)$. Then find $y$ when $x = 4$ in $y = 2x + 1$.

**18.** I can compare vertical and horizontal changes to find slope.

**Find the slope.**

Rise $-6$, run $3$

**19.** I can compare vertical and horizontal changes to find slope.

Find the slope between $(-2, 5)$ and $(2, -3)$.

**20.** I can compare vertical and horizontal changes to find slope.

Find $\Delta y$ and $\Delta x$ from $(1, 3)$ to $(4, 9)$.